

A4Q1

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Q1

(a)

```
df = read.csv("Stat341_S25_LinkedIn_Dataset.csv")
head(df)
```

##	AmpmEst	Comments	Reacts	Impressions	Members	Hashtags	Prompt	Picture	DOW
## 1	pm	0	16	876	410	Intermixed	No	No	Fri
## 2	pm	2	7	835	396	End	Yes	No	Mon
## 3	am	1	7	801	405	End	No	No	Wed
## 4	am	0	4	583	318	End	No	No	Wed
## 5	pm	11	16	4718	2829	End	Yes	Yes	Fri
## 6	pm	0	11	765	376	End	Yes	No	Mon

```
dim(df)
```

```
## [1] 24 9
```

```
x <- df$Reacts
y <- df$Impressions

r_result <- cor.test(x,y, method = "pearson")
r <- r_result$estimate

r
```

```
##          cor
## 0.5818502
```

```
n <- length(x)

t_obs <- r * sqrt(n-2) / sqrt(1-r^2)
p_val <- 2 * pt(-abs(t_obs), df = n-2)

p_val
```

```
##          cor
## 0.002858144
```

The resulting p-value is less than 0.05, providing sufficient evidence to reject the null hypothesis. This suggests that there is a statistically significant linear correlation between Impressions and Reacts in the dataset.

(b)

```
transforms <- list(
  identity = function(v) v,
  log = function(v) log(v + 1),
  sqrt = function(v) sqrt(v),
  inv = function(v) -1/(v + 1)
)

results <- data.frame(
  transformation = character(), r = numeric(), p = numeric()
)

for (name in names(transforms)) {
  tx <- transforms[[name]](x)
  ty <- transforms[[name]](y)
  r_test <- cor.test(tx,ty, method = "pearson")
  r_val <- r_test$estimate
  t_stat <- r_val * sqrt(n-2) / sqrt(1-r_val^2)
  p_val <- 2 * pt(-abs(t_stat), df = n-2)
  results <- rbind(results, data.frame(transformation = name, r=r_val, t=t_stat, p=p_val))
}

print(results)
```

```
##      transformation      r      t      p
## cor      identity 0.5818502 3.355629 2.858144e-03
## cor1         log 0.7467476 5.266151 2.772291e-05
## cor2         sqrt 0.6663445 4.191588 3.778853e-04
## cor3         inv 0.7581940 5.454121 1.767412e-05
```

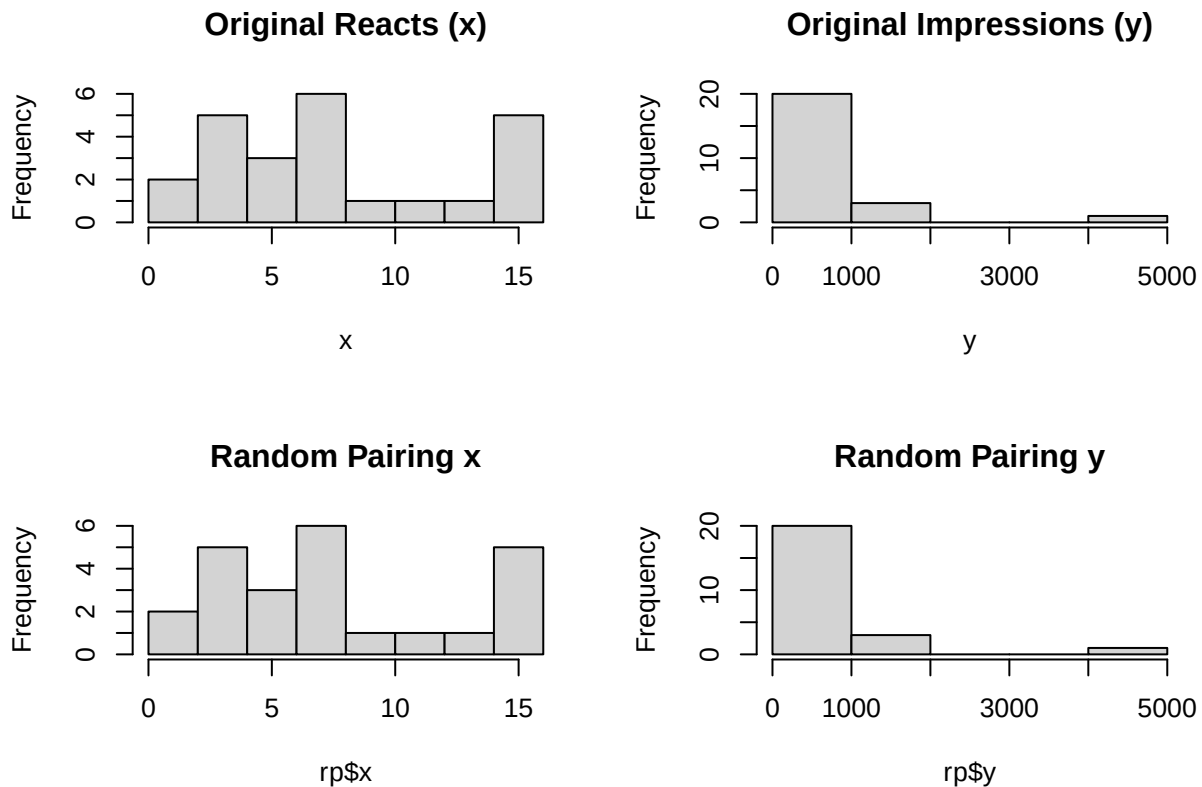
(c)

```
randomPairing = function(x,y)
{
  xrand = sample(x)
  yrand = sample(y)

  return(list(x=xrand, y=yrand))
}
```

```
rp <- randomPairing(x,y)
par(mfrow = c(2,2))

hist(x, main = "Original Reacts (x)")
hist(y, main = "Original Impressions (y)")
hist(rp$x, main = "Random Pairing x")
hist(rp$y, main = "Random Pairing y")
```



For Original React, the distribution right-skewed, with a cluster around smaller values and a few higher outliers, like around 15. In such case, $\log(x+1)$ or a square root transformation would be helpful in reducing skewness and stabilize variance.

For Original Impression, it is strongly right skew, with most of the values centered around 1000, and a few larger outliers, like 5000. This suggests a suitability of $\log(y+1)$ here to compress the large values and make the distribution more symmetric.

For both random pairing x and y, their distributions look nearly identical to the originals.

(d)

```
M <- 5000

t_scores <- replicate(M, {
```

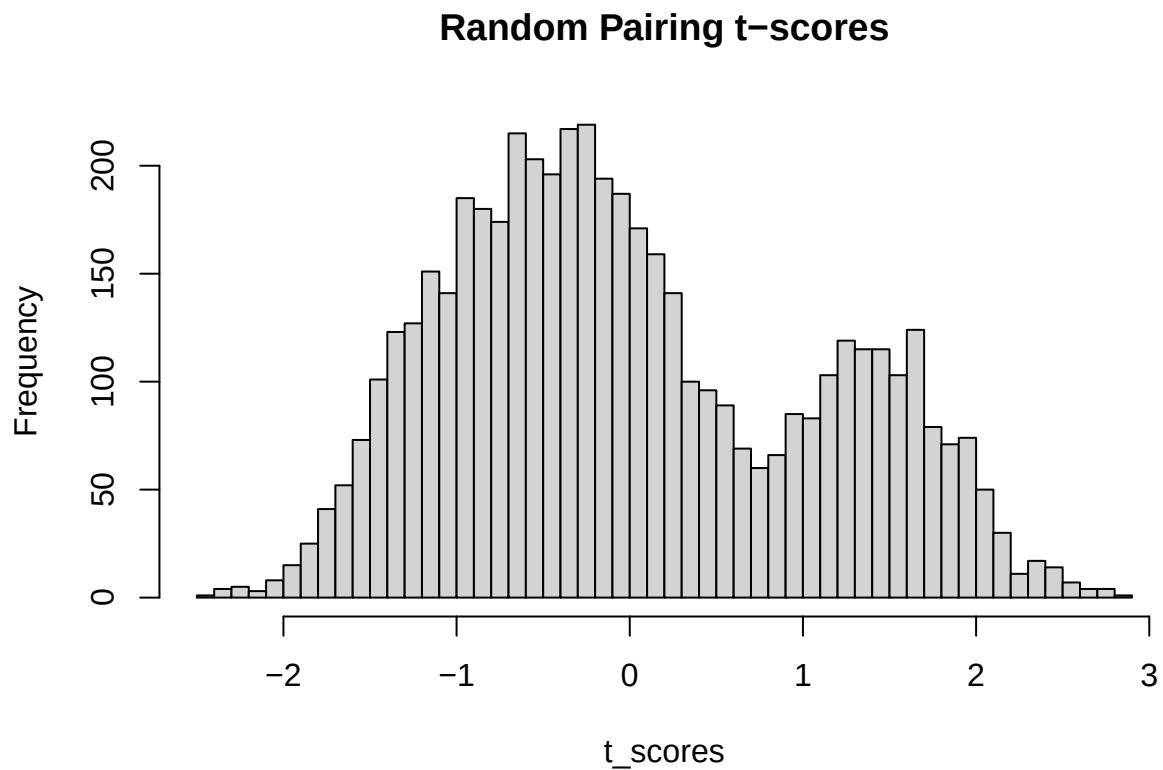
```

rp <- randomPairing(x, y)
rp_test <- cor.test(rp$x, rp$y, method = "pearson")
rp_val <- rp_test$estimate
rp_val * sqrt(n - 2) / sqrt(1 - rp_val^2)

})

hist(t_scores, breaks = 50, main = "Random Pairing t-scores")
abline(v = t_obs, col = "red", lwd = 3)

```



(e)

```

p_empirical <- mean(abs(t_scores) >= abs(t_obs))

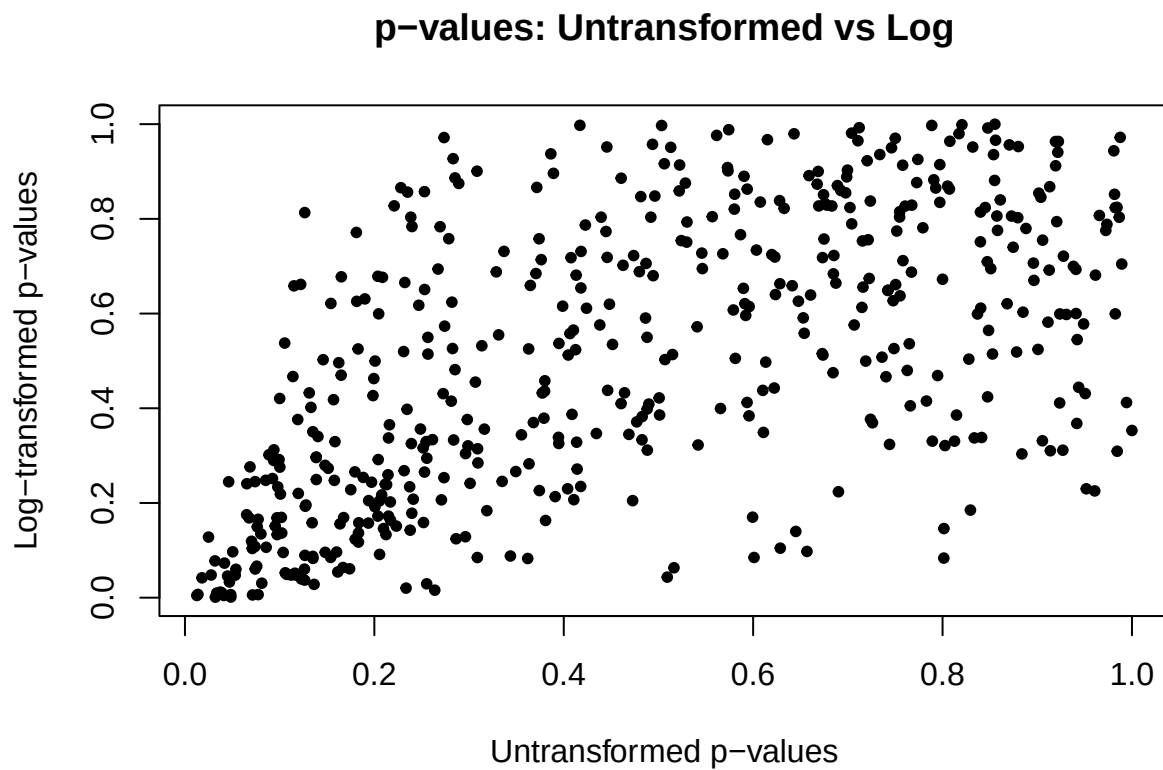
p_empirical

```

```
## [1] 0
```

(f)

```
get_pval <- function(x,y) {  
  r <- cor(x,y)  
  t <- r * sqrt(n - 2) / sqrt(1 - r^2)  
  2 * pt(-abs(t), df = n - 2)  
}  
  
pvals_matrix <- matrix(NA, nrow = M, ncol = 4)  
  
for (i in 1:M) {  
  rp <- randomPairing(x, y)  
  for (j in 1:4) {  
    tx <- transforms[[j]](rp$x)  
    ty <- transforms[[j]](rp$y)  
    pvals_matrix[i, j] <- get_pval(tx, ty)  
  }  
}  
  
plot(pvals_matrix[1:500, 1], pvals_matrix[1:500, 2],  
      xlab = "Untransformed p-values", ylab = "Log-transformed p-values",  
      main = "p-values: Untransformed vs Log", pch = 20)
```



(g)

```
observed_pvals <- sapply(transforms, function(f) {  
  tx <- f(x)  
  ty <- f(y)  
  get_pval(tx, ty)  
})  
  
observed_min <- min(observed_pvals)  
  
min_pvals_random <- apply(pvals_matrix, 1, min)  
p_adj <- mean(min_pvals_random <= observed_min)  
  
p_adj
```

```
## [1] 0
```

Among the 5000 random pairings, none of the transformations for each pairing produced a smaller minimum p-value, thereby the proportion is zero. This indicates a strong evidence against the null hypothesis of independence, even across multiple transformations.